

VULNERABILITY RESEARCH

INTRO TO OS

BOOTING PROCESS

STEP 1

POWER ON SELF TEST (POST)

Perform hardware checks to ensure all components are functionally correctly. This includes the CPU, memory, storage devices, and more

STEP 2

BIOS BOOT PROCESS

Following the successful completion of POST, the BIOS's attention shifts to the OS to consult the Bootable sequence. Once the MBR is found it loads the Bootloader

STEP 3

LOADING THE BOOTLOADER

The bootloader is loaded into memory. It's job is to load the os's kernel. In multi-boot systems, the bootloader may display a menu allowing the user to choose which OS to boot

STEP 4

KERNAL INITIALISATION

The kernal is then loaded into memory and starts initialising critical hardware components. The kernal will mount the root file system which contains essential files for the OS

STEP 8

USER AUTHENTICATION AND LOGIN

The user will then login using their preferred method. The OS checks this against a database and if authorised the users desktop environment is loaded

STEP 7

DISPLAYING THE LOGIN SCREEN

Once the system services are running, the GUI will load or the terminal login prompt usually displaying a login screen

STEP 6

STARTING SYSTEM SERVICES/DAEMONS

System services (windows)/ Daemons (unix systems) now start. They prepare the system environment and start necessary background tasks required for the OS to function smoothly.

STEP 5

LOADING DEVICE DRIVERS

The OS will load the device drivers which allows the OS to communicate with hardware devices, such as graphic cards, network card, harddrive etc